Formlabs and EOS are also into printing surgical instruments.

Sowing the seeds of innovation in India. Very recently, the Indian Institute of Science (IISc), Bengaluru, partnered with CELLINK—a global leader in 3D bioprinting—to establish a Centre of Excellence (CoE) for 3D bioprinting in India. The CoE will be situated at the Centre for Bio Systems Science and Engineering (BSSE) at IISc and will house several CELLLINK 3D bioprinters.

The Centre will provide researchers with quick access to 3D bioprinting systems, accelerating their efforts. It will also spearhead research and training initiatives related to the applications of 3D bioprinting in fields like tissue engineering, drug discovery, material science, and regenerative/personalised medicine. There will be a special focus on the heart, bone cartilage, and cancer.

Light and strong parts for cars and crafts that fly

The automotive and aerospace industries have also recognised the benefits of 3D printing and started adopting the tech, especially for low-volume, customised parts.

A Ford plant near Cincinnati saved millions of dollars by 3D printing a couple of expensive and hard-to-get spares, like pucks and bumpers. A General Motors facility in Tennessee also experienced significant savings by replacing conventional tooling components with 3D-printed alternatives made in-house. The 3D-printed tooling had better functionality despite being developed faster and at a lower cost.

The kind of design freedom that 3D printing gives is a boon for carmakers, especially in the luxury segment. Carmakers like Bentley are increasingly turning to 3D printing to customise their offerings and give customers exactly what they want.

With 3D printing, auto-makers can also ensure a proper supply of parts to customers, even those that went out of production years ago. This is a dream-come-true for vintage car and bike collectors, who are now able to restore and preserve their old models in working condition. Service centres in remote areas could also benefit by the ability to 3D-print spares instead of waiting for them to arrive from faraway warehouses.

3D printing is a real disruptor in the racing world, where aerodynamically-optimised parts with the right materials and designs can clinch a win. It is not surprising that most racing specialists today have partnered with 3D printing firms—Ducati Corse with Roboze, Alfa Romeo with Additive Industries, EOS with the Camozzi Group, NASCAR with Stratasys, and so on.

Several parts, ranging from brake ducts, brackets and supports, to seals and cockpit ventilation systems, are being 3D-printed today. Material innovation is also at its peak in this space—new composites are giving these cars and bikes the right mix of strength, weight, and temperature tolerance.

If these factors are so important for automobiles, you can easily imagine how critical they would be for aircrafts and spacecrafts. No wonder then that the aerospace industry is an eager taker for this tech. Additive manufacturing is being used to print a lot of stuff, from jigs and fixtures to surrogates, mounting brackets and detailed prototypes.

The tech is helping engineers make lighter and more efficient engines and turbine parts. The ability to print replacement parts reduces downtime and also extends the life of aircrafts, as even obsolete parts can be upgraded to current standards and 3D-printed.

A Dassault Systèmes report recalls how Satair, an aircraft component and service company based in Denmark and a subsidiary of Airbus, provided one of its airline customers in the US with a 3D-printed spare part in 2020. The part was no longer being produced by the original supplier. Redesigning it and producing it using conventional methods was an expensive proposition. So, using a new certification process, Satair was able to recertify the former cast part within five weeks and adapt it to titanium, a material well-suited for additive manufacturing and for use in aircrafts. The part was then 3D-printed and supplied to the obviously happy customer!

Interestingly, additive manufacturing is used extensively to print parts of cabin interiors like seats and handles—in an effort to reduce weight while maintaining aesthetics. Reducing weight helps reduce air drag, saves fuel, and correspondingly carbon dioxide emissions too.

The Peekay Group in association with Bengaluru Airport City Limited (BACL) has set up a new state-of-the-art 3D printing facility at the Airport City. It has a production centre and an experience zone, where individuals and students can come and try their hand at 3D printing. It will also incubate worthy projects.

The Peekay Group also plans to install a 3D metal printing unit to cater to the aerospace industry. They hope to add more 3D printing processes, so students can get more experience in environment-friendly, design-oriented manufacturing, while technicians can also upskill themselves.

Your building, as you like it, as fast as you want it

In April 2021, Tvasta Construction, a startup incubated at IIT Madras, built a 56 square metre (600 sq ft) house on the university campus. The quaint house has a hall, a kitchen, and a bedroom. Sounds normal? Well, what makes it special is that it was built within five days using concrete 3D printing technology.

More recently, the Indian Army's Military Engineering Services (MES) partnered with Tvasta to 3D-print two houses for soldiers at the South-Western Air Command, Gandhinagar, within a month's time. The 65 square